TEACHING SCRIPT: Computer Handling and Care

Topic: Hardware Care — Cleaning Dust, Keyboard, Mouse, and Monitor

Form One — Computer Science | Reference: Student's Book, Form One (pages 83–87)

This is a complete script you can read and explain to students from beginning to end. Under every point from the book, there is a simple explanation and everyday examples to help low-level learners understand.

OPENING (What to say to start the lesson)

"Good morning, class. Today we are moving to a new sub-topic called **Computer Handling and Care**. Specifically, we are going to talk about **Hardware Care** — how we look after the physical parts of a computer: the case, the keyboard, the mouse, and the monitor (screen).

Before we start, let me ask you something simple: Do you sweep your house? Do you wash your clothes? Do you bathe every day? Why do you do these things?"

(Let students answer — expect things like "to stay clean", "to avoid sickness", "to look smart")

"Exactly! A computer is the same. It also needs to be kept clean and cared for, or it will get 'sick' — meaning it will start having problems, running slowly, becoming noisy, or even getting completely damaged. Today we will learn **why** dust and dirt are dangerous to a computer, and **how** to clean the different parts safely."

PART 1: Aspects of Computer Handling (Introduction)

🗐 **From the book:** Computer handling attributes include hardware care, software care, and safe handling practices.

🗣 **Simple explanation:** "Class, when we say 'handling' a computer, we mean everything we do with it — from the moment we touch it, use it, move it, to how we protect it. There are three main areas: 1. **Hardware care** – taking care of the physical parts (screen, keyboard, mouse, casing). 2. **Software care** – taking care of the programs inside it (not installing bad software, updating it, not opening dangerous files). 3. **Safe handling practices** – the correct behaviour when using and moving the computer (e.g., not eating near it, not dropping it, switching it off properly).

Today's lesson focuses on the first one: **Hardware care**."

Example to use: "Think of a computer like a motorcycle (bodaboda). Hardware care is like taking care of the engine and the body of the bike — washing it, checking the tyres, oiling the chain. If you don't do

this, the bike will break down quickly, even if the rider (the software/driver) is very good."

PART 2: What is Hardware Care?

📖 **From the book:** Hardware care involves cleanliness, temperature control, ventilation, power management, relocation of a computer, and controlling the effect of liquid on computer hardware.

🗣 **Simple explanation:** "Hardware care is not just about wiping dust. It is a group of habits: - **Cleanliness** – keeping it free from dust and dirt. - **Temperature control** – not letting the computer become too hot. - **Ventilation** – making sure air can move in and out of the computer freely (like how a room needs open windows). - **Power management** – using electricity correctly (not turning on/off carelessly, using a stabilizer where possible). - **Relocation** – carrying and moving the computer carefully from one place to another. - **Controlling liquid** – keeping water, juice, tea, and other liquids far away from the computer.

Today we will go deeper into **cleanliness** — because this is the part most students forget, yet it causes the most damage."

PART 3: Why Do We Need to Clean the Computer? (Cleaning)

📖 **From the book:** Computers should be kept in a clean, conducive environment to function properly. It is advisable to regularly clean the hardware and peripherals (keyboard, mouse, monitor) to remove dust and dirt using suitable cleaning equipment, in order to minimise harm to the components.

🗣 **Simple explanation:** "A computer is like a living creature that 'breathes' air to cool itself down. If the air it breathes is full of dust, just like a person breathing in a dusty room, it will start to struggle. That's why we must clean it **regularly**, using the **correct tools** — not just any cloth or any liquid, because the wrong material can damage sensitive parts."

Example: "Imagine you never sweep your classroom floor for six months. Dust will pile up in the corners, under the desks, and even on the chalkboard. Slowly, the classroom becomes uncomfortable, unhealthy, and hard to use. That is exactly what happens **inside** a computer when we don't clean it — except the 'classroom' is small and the dust gets trapped in tiny fans and vents where we cannot even see it."

PART 4: The Danger of Dust — Removing Dust

📖 **From the book:** Dust is a common environmental impurity that accumulates inside a computer over time if the system is not adequately protected or regularly maintained.

🗣 **Simple explanation:** "Now let's understand **exactly** what dust does inside a computer. Dust doesn't just sit there quietly — it actively causes six serious problems. I want you to write these down and I will explain each one with an example."

(i) Reduced Cooling Efficiency

📘 **Book says:** Dust can block air vents, cooling fans, and heat sinks. This blocks airflow, disrupts cooling, and raises the temperature inside the system, which can make the CPU and GPU overheat, causing slow performance or hardware failure.

🗣️ **Simple explanation + Example:** "A computer has small fans inside it, just like the fan you use at home to cool yourself when it's hot. Now imagine putting a wet, dusty cloth over your fan at home — the wind can no longer pass through properly, and the room stays hot. That is what dust does to a computer's fan: it blocks the air from moving, so the important parts inside (the 'brain' of the computer, called the CPU, and the 'graphics brain', called the GPU) become too hot, just like a person getting a fever."

(ii) Increased Noise

📘 **Book says:** When cooling fans are clogged with dust, they must work harder, spinning faster, which increases fan noise — and this can also be a sign of hardware problems.

🗣️ **Simple explanation + Example:** "Have you ever seen someone carrying a heavy bucket of water uphill? They breathe loudly and struggle, right? That is exactly what a dusty fan does — because the dust is blocking its path, the fan has to spin harder and faster to push air through, and that is why you start hearing a loud, annoying noise from the computer. A noisy computer fan is like a person breathing heavily after running — it's a sign of stress."

(iii) System Instability and Crashes

📘 **Book says:** Overheating caused by dust can lead to system instability and unexpected crashes — the computer may freeze or shut down suddenly.

🗣️ **Simple explanation + Example:** "When your body temperature becomes too high (a very high fever), you may faint or collapse suddenly. A computer behaves the same way. When it overheats because of dust, it may suddenly freeze (stop responding to your clicks) or shut down completely on its own, even in the middle of your work — and you can lose everything you were doing!"

(iv) Shorter Component Lifespan

📘 **Book says:** Excessive heat due to dust causes stress on internal components, reducing their lifespan — processors, graphics cards, and other hardware wear out faster.

🗣️ **Simple explanation + Example:** "Think about a person who works in the hot sun every single day without rest or shade — their body wears out faster than someone who rests in a cool place. In the same way, computer parts that are constantly overheating because of dust get 'tired' faster and die before their time. A computer that should last you 6 years might only last 2 or 3 years because of dust damage."

(v) Impaired Performance

📘 **Book says:** Dust can affect the electrical connections between components, causing intermittent connectivity issues or data transfer errors, resulting in reduced performance and slower response times.

🗣 **Simple explanation + Example:** “Inside a computer, tiny electrical connections pass information between parts, like small roads connecting different towns. When dust settles on these connection points, it’s like potholes and mud blocking a road — cars (information) move slower or get stuck. That’s why a dusty computer becomes slow: opening a file takes longer, a video may lag, or the computer may ‘hang’.”

(vi) Fire Hazard

📖 **Book says:** While rare, a significant buildup of dust can pose a fire hazard, especially near electrical components, since dust is flammable and can catch fire if exposed to a spark or overheated component.

🗣 **Simple explanation + Example:** “This is the most serious danger. Dust is actually a material that can burn, just like dry grass in the dry season. If dust builds up near an electrical part that sparks or becomes extremely hot, it can catch fire. It is rare, but it has happened, especially in old computers that were never cleaned for years. This is why cleaning a computer is not just about making it look nice — it can even protect us from danger.”

🗣 **Quick class recap for Part 4:** “So class, let’s summarise the 6 dangers of dust using one simple sentence each: 1. Blocks air → computer gets hot (like blocking a fan). 2. Fan works harder → makes noise (like a person struggling to breathe). 3. Overheating → computer freezes/shuts down (like a person fainting from fever). 4. Constant heat → parts die early (like working non-stop in the sun). 5. Dirty connections → computer becomes slow (like a muddy road). 6. Too much dust → can even cause fire (like dry grass catching fire).”

PART 5: How to Clean Dust from a Desktop Computer (Step by Step)

📖 **From the book:** Before cleaning, gather materials: soft cloths, an electric air blower, isopropyl alcohol (ethanol or hydrogen peroxide), and a soft brush.

🗣 **Simple explanation:** “Class, before we clean anything, we must first prepare our tools — just like a doctor prepares equipment before treating a patient. You cannot clean a computer using just any cloth or any liquid from home, because some materials can damage the delicate parts inside. The book tells us we need: a soft cloth, an air blower, a special cleaning alcohol, and a soft brush.”

Now let’s go through the steps, one by one:

Step (i): Gather your materials Soft cloths, electric air blower, isopropyl alcohol, and a soft brush. *Simple tip:* “Prepare everything before you start — like a nurse arranging tools on a tray before an operation.”

Step (ii): Carefully remove the side panel of the computer case Some cases have screws; others have latches or buttons. *Simple tip:* “Go slowly. Don’t force it. If there are screws, keep them somewhere safe so you don’t lose them.”

Step (iii): Observe the interior of the computer Look for areas with heavy dust buildup — especially the cooling fans, heat sinks, and other dusty components. *Simple tip:* “This is like a doctor first examining a patient before giving treatment. Look before you touch.”

Step (iv): Use an electric air blower to blow away loose dust Hold the fans still to prevent them from spinning while blowing air, to avoid damage. *Simple tip:* "⚠ This is very important — if a fan spins too fast while you are blowing air into it, it can generate electricity backwards and damage the computer. Always hold the fan blades still with one hand while blowing with the other."

Step (v): Use a soft brush to gently loosen the dust Avoid too much force, since some parts are delicate. *Simple tip:* "Be as gentle as when you are cleaning a baby's skin — soft and careful, not rough."

Step (vi): Use a soft cloth to wipe surfaces inside the case You may lightly dampen the cloth with isopropyl alcohol for tougher cleaning — but never let liquid touch sensitive electronic parts. *Simple tip:* "⚠ 'Lightly damp' means just a little bit wet, not soaked. If you squeeze the cloth and water drips out, it is too wet — that is dangerous for the computer."

Step (vii): Pay special attention to the cooling fans and vents These affect airflow and cooling the most, so clean them thoroughly. *Simple tip:* "Remember our earlier lesson — dust on the fan is the number one enemy of the computer. Give it the most attention."

Step (viii): Clean peripheral devices (keyboard and mouse) Blow away loose dirt with compressed air, and wipe with a slightly damp cloth if needed.

Step (ix): Carefully reassemble the computer Make sure all components are secure, reconnect all cables and peripherals, then switch it on and check that everything works correctly. *Simple tip:* "This is like a mechanic closing the bonnet of a car after servicing the engine. Make sure nothing is left loose or disconnected before you switch it on."

🗐 **From the book — Recommended cleaning schedule:** It is recommended to clean a computer once every **three to six months**, depending on the environment.

🗣 **Simple explanation:** "If you live or study in a very dusty area — like near a road, in a windy area, or where there's construction — clean more often, maybe every 3 months. If the environment is cleaner, every 6 months is fine. Think of it like cutting your hair — dusty environment means you 'cut' the dust more often."

⚠ Safety Precautions When Cleaning (Very Important — repeat this to students twice)

🗐 **From the book:** 1. Never use a vacuum cleaner to clean the interior of a computer — it can create static electricity and damage sensitive components. 2. Do not use water or household cleaning solutions on internal components. 3. Always ensure the computer is unplugged and disconnected from the power source before cleaning.

🗣 **Simple explanation + Example:** "Class, these three rules can save you from destroying an entire computer, so listen carefully:

1. **No vacuum cleaners inside the computer.** A vacuum cleaner builds up static electricity — an invisible electrical charge — similar to when you rub a balloon on your hair and it sticks. That static charge can jump onto the computer chips and destroy them instantly, even though you don't feel it happening.

2. **No water or soap on the inside.** Water conducts electricity. If water touches an active electrical circuit, it can cause a short circuit — just like when a wet switch at home can shock you or spark. Water and electronics do not mix.

3. **Always unplug first.** This is the golden rule of safety — for the computer's protection AND for your own protection from electric shock. It's the same reason your teacher tells you to switch off the lights before changing a bulb at home."

PART 6: Cleaning the Keyboard

📖 **From the book:** A keyboard allows dust and particles to enter and accumulate over time. Periodic cleaning prolongs its lifespan and prevents malfunction.

🗣 **Simple explanation:** "Look at your keyboard, class — do you see the small spaces between the keys? Those spaces are like small doors that allow dust, crumbs, and dirt to fall inside and get stuck. Over time, this can make some keys stop working properly, or become sticky when pressed."

Steps for cleaning a keyboard:

(i) Clean the keyboard regularly to remove dust, dirt, and debris between the keys.

(ii) Use a blower to blow away particles, or gently shake the keyboard upside down. *Simple tip:* "Turn the keyboard over gently like you are shaking sand out of your shoe — you'll be surprised how much dirt falls out!"

(iii) Wipe the surface and keys with a soft cloth slightly dampened with water or a mild cleaning solution.

(iv) Avoid harsh chemicals or excessive moisture, as they may damage the keyboard.

(v) Use a keyboard cover or silicone skin to protect it from dust, dirt, and spills. *Simple tip:* "This is like wearing a raincoat — it protects you from the rain before it even touches your skin. A keyboard cover protects the keys before dust even reaches them."

(vi) Refrain from eating or drinking near the keyboard to prevent spills and crumbs from getting inside.

🗣 **Real-life example to give students:** "Imagine you are eating chapati and drinking juice right next to your keyboard while typing an assignment. A crumb falls between the keys, or the juice spills — now some letters may stop working, or worse, the liquid can reach the circuit board underneath and completely destroy the keyboard. This is why the book warns us: **never eat or drink near the keyboard**. It's such a common mistake that many computers get damaged this way, not from old age, but from a simple spilled drink."

PART 7: Cleaning an Optical Mouse

📖 **From the book:** To clean an optical mouse, follow these general steps.

🗣 **Simple explanation (introduction):** "Now let's talk about the mouse — the small device you use to point and click. Underneath the mouse, there is a small sensor (like a tiny 'eye') that reads the surface and tells the computer where to move the pointer. If that 'eye' gets dirty, the mouse pointer may move roughly, jump around, or stop responding smoothly."

Steps for cleaning an optical mouse:

(i) Disconnect the mouse — unplug it from the computer or device. *Simple tip:* "Just like turning off a tap before fixing a pipe — always disconnect power/connection before cleaning."

(ii) Turn the mouse upside down — this lets you access the bottom, where the optical sensor and pads are located.

(iii) Inspect and clean the pads — check for accumulated dirt or debris (too much debris makes the mouse feel stiff when moving). Use a toothpick, a clean lint-free cloth, or a cotton cloth dampened with mild soap and water to gently wipe the pads. *Simple tip: "*A toothpick is perfect for reaching into tiny corners — like using a small stick to remove something stuck between your teeth."

(iv) Clean the optical sensor — usually located at the centre of the mouse's bottom surface. Use a clean, lint-free cloth or a cotton swab dampened with mild soap and water to gently wipe the sensor. Be careful not to press too hard or let liquid get inside the mouse. *Simple tip:* "⚠ The sensor is like the 'eye' of the mouse — you must clean it as gently as you would clean your own eye. Never scrub it hard."

(v) Remove accumulated lint and debris — use a can of compressed air to blow away dust between the buttons. Hold the can upright and use short bursts of air to avoid spraying liquid. *Simple tip:* "Holding the can upside down can cause it to spray cold liquid instead of air, which can damage the mouse — always keep the can standing straight up."

(vi) Allow the mouse to dry — let it air dry completely before reconnecting it to the computer. *Simple tip:* "Just like you don't wear a shirt that is still wet from washing, don't plug in a mouse that is still damp from cleaning — wait for it to dry first."

PART 8: Cleaning a Computer Monitor (Screen)

📄 **From the book:** To clean a computer monitor, follow these general steps.

🗣 **Simple explanation (introduction):** "The monitor is the screen we look at every time we use the computer. It is also one of the most delicate parts — the screen surface can scratch or get damaged easily if we use the wrong material or too much pressure."

Steps for cleaning a computer monitor:

1. Turn off the monitor — switch it off and unplug it from the power source before cleaning, for safety and to prevent damage.

2. Use a microfiber cloth — a soft, lint-free cloth that will not scratch the monitor's surface. *Simple tip: "*A microfiber cloth is the same soft cloth used to clean eyeglasses — very gentle. Avoid rough fabrics like an old t-shirt or a kitenge cloth, which may scratch the screen."

3. Dampen the cloth — moisten the microfiber cloth with distilled water or a cleaning solution made specifically for monitors. **Do not spray liquid directly onto the screen** — it can seep inside and cause damage. Use only a small amount of liquid on the cloth. *Simple tip:* "⚠ This is one of the most important rules: never spray water directly onto the screen like you spray a plant. Spray it onto the cloth first, then wipe the screen with the cloth."

4. Wipe the screen — gently wipe in a circular motion or from top to bottom. Avoid excessive pressure, as it can damage the screen. Pay extra attention to areas with smudges or fingerprints. *Simple tip:* "Wipe the screen the same gentle way you would wipe a newborn baby's face — softly, without pressing hard."

5. Dry the screen — use a dry microfiber cloth to remove any leftover moisture before turning the monitor back on. *Simple tip:* "Make sure the screen is completely dry, just like you wouldn't switch on a wet phone — moisture and electricity together are dangerous."

🗣 **Important warning to repeat to students:** "Class, remember: **avoid using paper towels, tissues, or rough fabrics** on a screen. These materials look soft, but they actually have tiny rough fibres that can leave permanent scratch marks on your monitor. Always use a proper microfiber cloth."

CLOSING SUMMARY — "Methods I Recommend You Emphasise Most" (End of Lesson Recap)

Read this out loud to students as a final summary before ending the lesson:

✅ **Clean your computer regularly** — every 3 to 6 months, depending on how dusty your environment is.

✅ **Always unplug the computer before cleaning it** — for your safety and the computer's safety.

✅ **Never use water directly, a vacuum cleaner, or rough cloths** on any computer part.

✅ **Use the right tools**: soft brush, air blower, microfiber cloth, and a cloth only lightly dampened when needed.

✅ **Never eat or drink near the keyboard** — spills are one of the most common causes of computer damage.

✅ **Handle the mouse and screen gently** — they contain delicate sensors and surfaces that scratch or break easily.

✅ **Dust is dangerous** — it causes overheating, noise, slow performance, shorter lifespan, and in rare cases, even fire.

Closing line for the teacher to say: "So class, remember — a computer is like a living helper in your studies and future work. Just as you take care of your own body through cleanliness, you must also take care of your computer. A well-cared-for computer will serve you for many years; a neglected, dusty computer will fail you when you need it most — maybe even during an important exam or assignment deadline. Are there any questions before we move to the next sub-topic?"

End of script. Reference: Computer Science, Student's Book, Form One (WazaElimu.com), pages 83–87.